

INT422 IT Project Management

Unit II

Project Roles

Lecture 14

Outline

- Team structures
- Project planning
- Work breakdown structure
- Communication Management
- Dependencies
- Schedule
- Project Management Tools

Project Roles

- Planner
- Analyst
- Designer
- Programmer
- Tester
- Maintainer
- Trainer
- Document Editor
- Web Master
- Configuration Manager
- Group leader
- Liaison
- Minute Taker
- Project Manager

Project Roles

- Management roles
 - Organization and execution of the project within constraints. Examples: project manager, team leader.
- Development roles
 - Specification, design and construction of subsystems. Examples: Analyst, system architect, implementer.
- Cross functional roles
 - Coordination of more than one team. Example: API Engineer, configuration manager, tester
- Consultant roles
 - Support in areas where the project participants lack expertise. Examples: End user, client, application domain specialist (problem domain), technical consultant (solution domain).
- Promoter roles
 - Promote change through an organization.

Promoter Roles

- Promoters are self appointed individuals who identify themselves with the outcome of the project.
- They are member of the corporate organization and may not necessarily be directly involved with the project. Instead, they are interfaces to the rest of the corporate organization.
- Because of the power, knowledge of technology, or familiarity with the project's processes, they are able to promote and push specific changes through the organization.

Power Promoter

- Also called executive champion
- Pushes the change through the existing organizational hierarchy.
 - not necessarily at the top of the organization, but must have protection from top level management, otherwise project opponents might be able to prevent the success of the project.
- **Tasks:**
 - constantly identify difficulties, resolve issues, and communicate with the project members, especially with the developers.
- **Example at project level:** Project Manager.
- **Example at corporate level:** Chief Executive Officer (CEO).

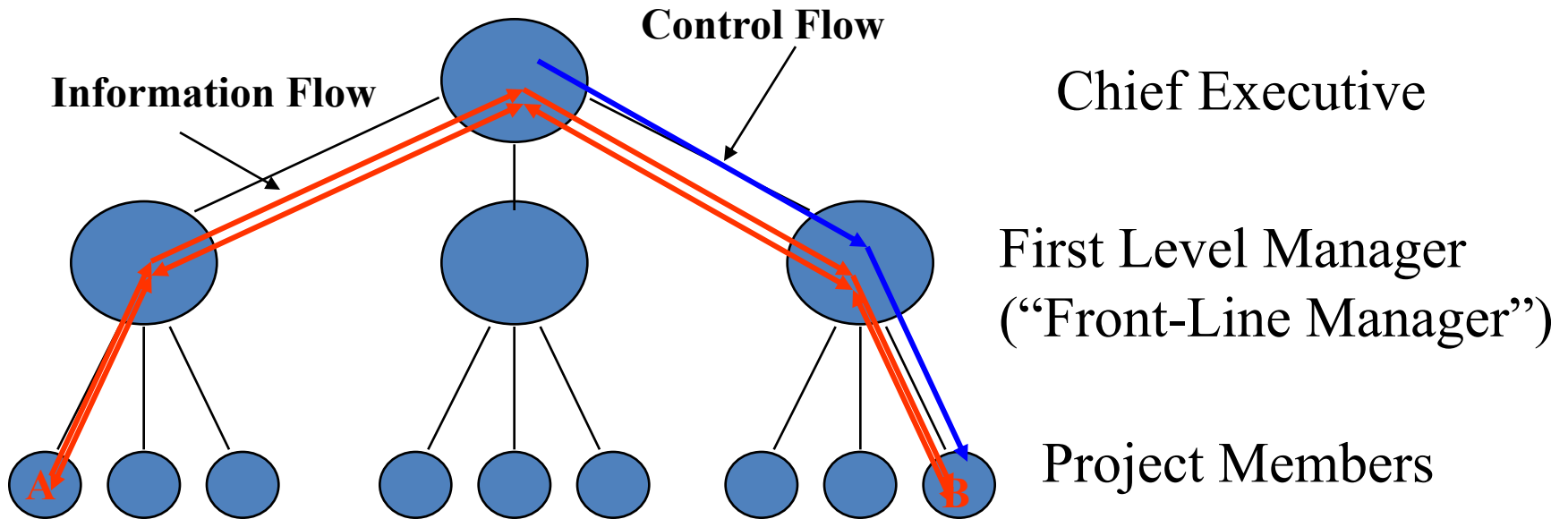
Knowledge Promoter

- Also called the technologist,
- Promotes change arising in the application domain or the solution domain. Usually associated with the power promoter.
- **Tasks:** Acquire information iteratively, understand the benefits and limitations of new technologies, and argue its adoption with the other developers.
- **Example at project level:** System architect.
 - Reports to project manager
 - Does not have any direct subordinate in the reporting hierarchy
 - Has final say over all technical decisions in the system.
- **Example at corporate level:** Chief Technical Officer (CTO).

Process Promoter

- The process promoter has intimate knowledge of the projects processes and procedures.
- The process promoter is in constant interaction with the power promoter to get consensus on the overall goals.
- **Tasks:** Bridge between the power and knowledge promoters, who often do not speak or understand the same language.
- **Example at project level:** Development lead. Responsible for the administrative aspects of a project, including planning, milestones definition, budgeting and communication infrastructure.
- **Example at corporate level:** Chief Information Officer (CIO)

Project Management: Hierarchical Project Organization

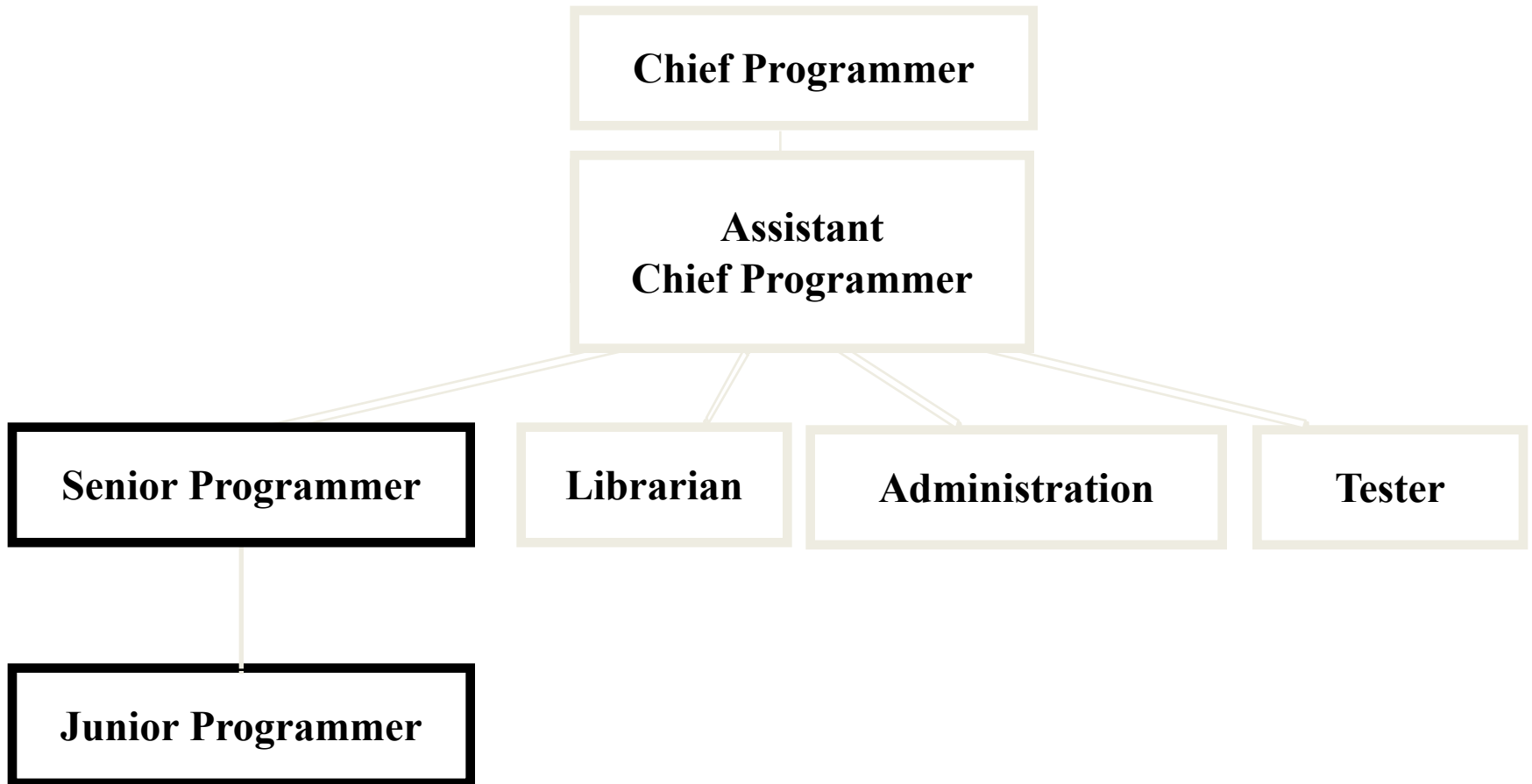


A wants to talk to B: Complicated Information Flow

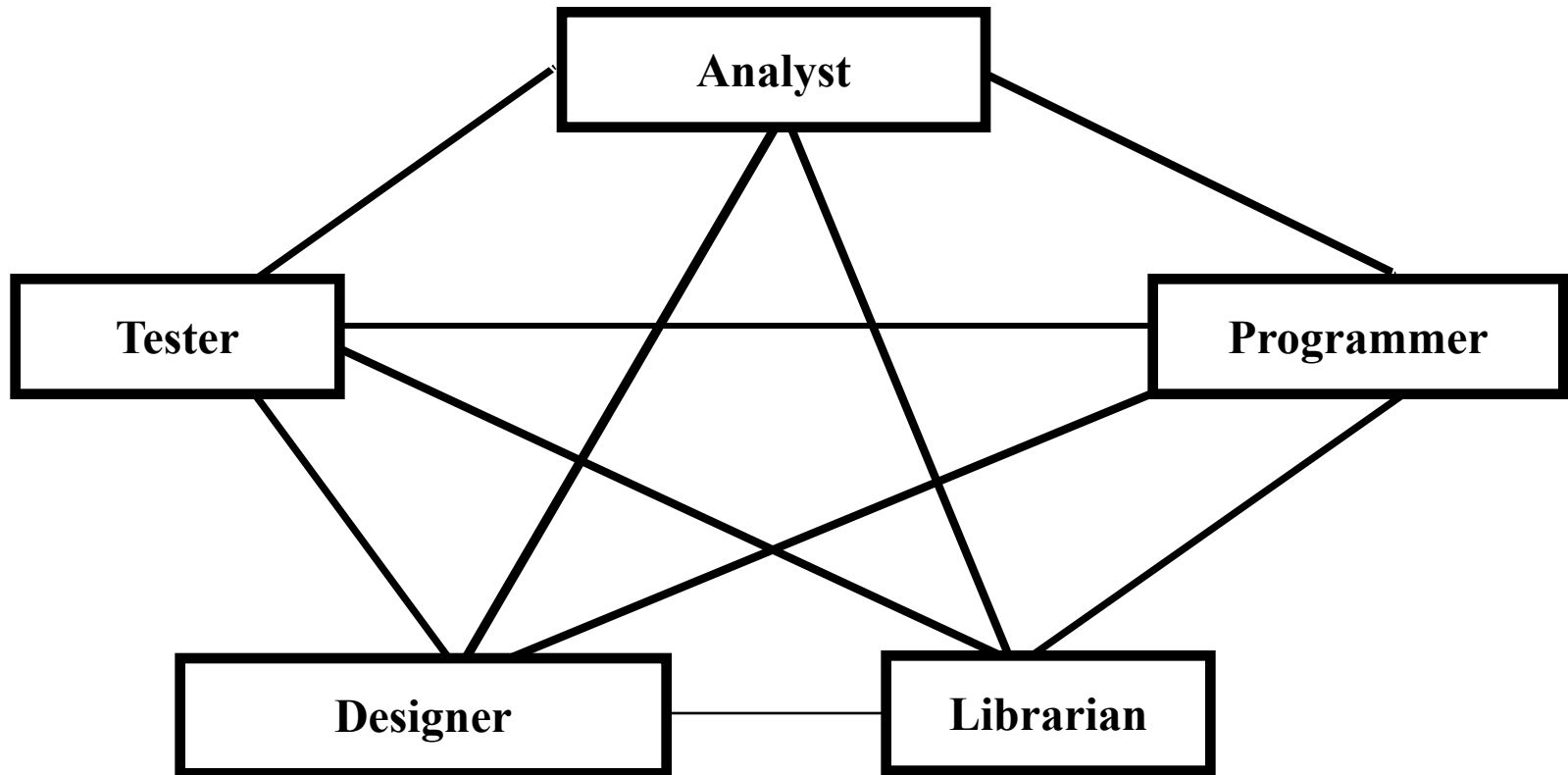
A wants to make sure B does a certain change: Complicated Controlflow

Basis of organization:
Complicated information and control flow
across hierarchical boundaries

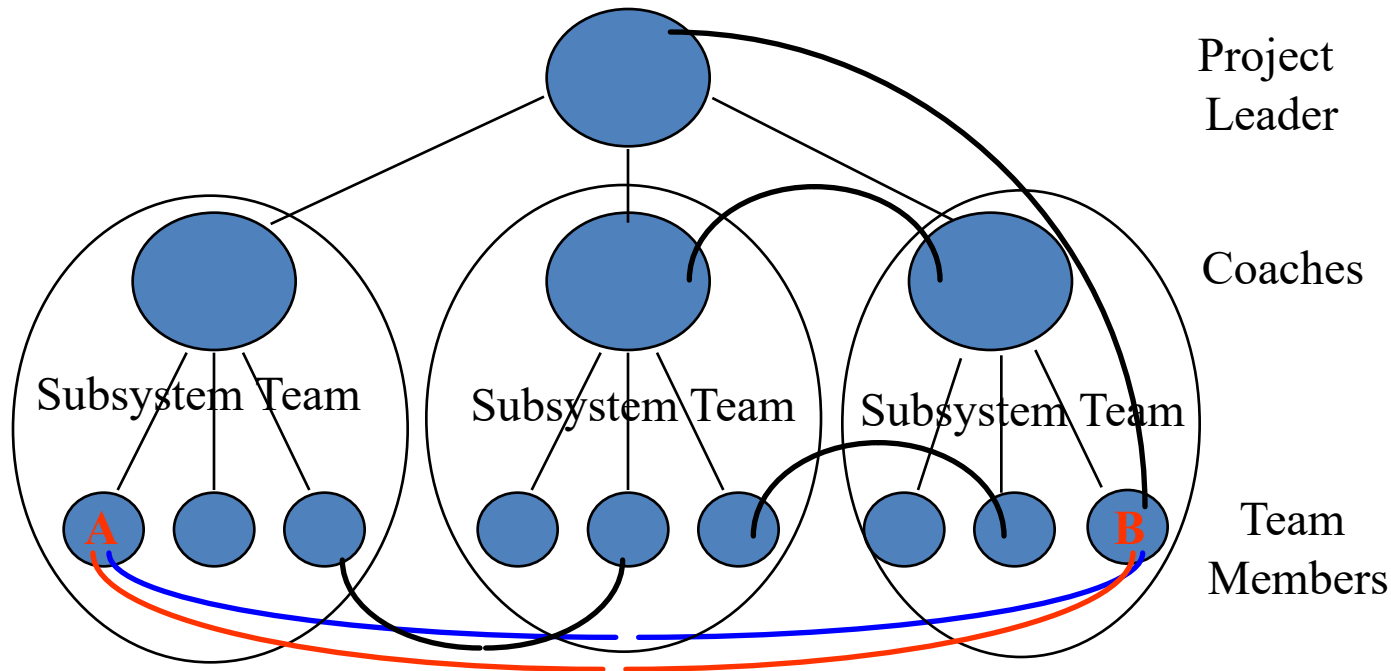
Example of Hierarchical Organization: Chief Programmer Team



Another Project Organization: Egoless Programming Team (Weinberg)



Project-Based Project Organization



A wants to talk to B: Communication Flow

A wants to make sure B does a certain change: Decision Flow

Basis of organization:
Nonlinear information flow across dynamically formed units

Associations in organizational structures

- Reporting association:
 - Used for reporting status information
- Decision association
 - Used for propagating decisions
- Communication association
 - Used for exchanging information needed for decisions (e.g., requirements, design models, issues).

Observations on Management Structures

- Hierarchical structures
 - “Reports”, “Decides” and “Communicates-With” all mapped on the same association
 - Do not work well with iterative and incremental software development process
 - Manager is not necessarily always right
- Project-based structures
 - “Reports”, “Decides” and “Communicates-With” are different associations
 - Cut down on bureaucracy reduces development time
 - Decisions are expected to be made at each level
 - Hard to manage

Hierarchical Structure

- Projects with high degree of certainty, stability, uniformity and repetition.
 - Requires little communication
 - Role definitions are clear
- When?
 - The more people on the project, the more need for a formal structure
 - Customer might insist that the test team be independent from the design team
 - Project manager insists on a previously successful structure

Project-Based Structure

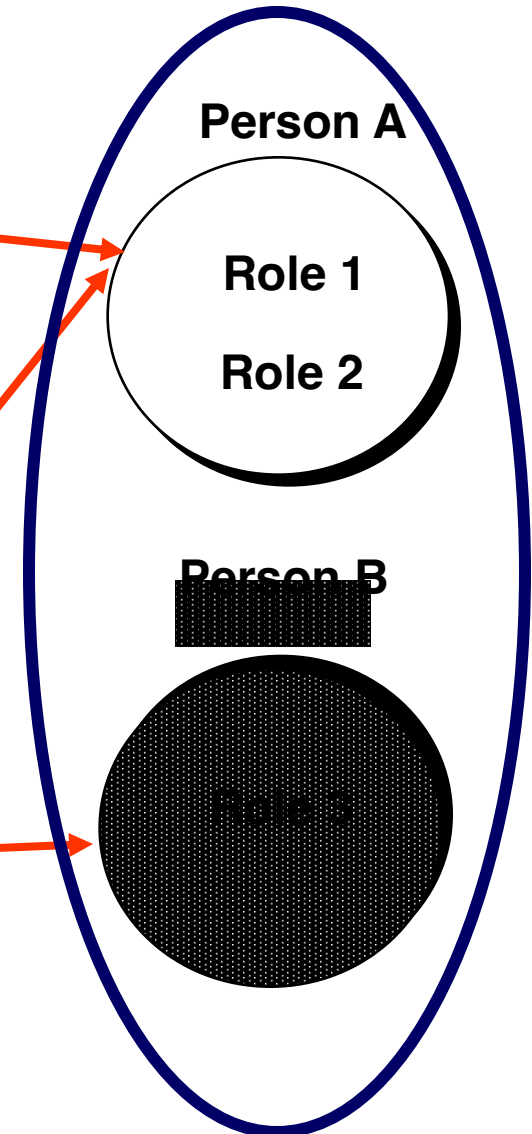
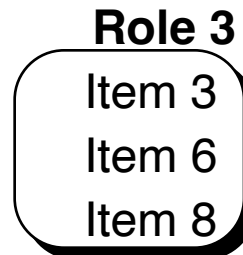
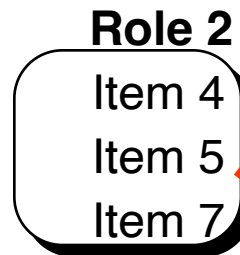
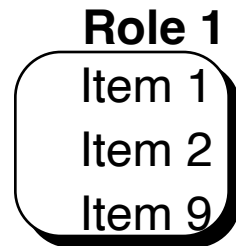
- Project with degree of uncertainty
 - Open communication needed among members
 - Roles are defined on project basis
- When?
 - Requirements change during development
 - New technology develops during project

Assigning Responsibilities To People

Team A

“To Do” List for the Project

- Item 1
- Item 2
- Item 3
- Item 4
- Item 5
- Item 6
- Item 7
- Item 8
- Item 9



Possible Mappings of ToDos to People

- One-to-One
 - Ideal but often not worth to be called a project
- Many-to-Few
 - Each project member assumes several roles ("hats")
 - Danger of overcommittment
 - Need for load balancing
- Many-to-"Too-Many"
 - Some people don't have significant roles
 - Bystanders
 - Loosing touch with project

Team Formation

- Top level Design
 - “Rough” Subsystem Decomposition (before requirements analysis)
 - Done during Predevelopment phase
- Team Formation done after Top Level Design
- Heuristics:
 - One team for each subsystem
 - One cross-functional task per team
 - 5-7 members per team
- Be prepared to iterate the team formation after system design when the subsystem decomposition is baselined

Project Roles: Coach

- Listen to gripes from individual teams
- Review weekly team reports
- Attend weekly project meetings
- Schedule and prepare meetings with client
- Insist that guidelines are followed
- Assign presentations (in-class project meetings, client review, client acceptance test)
- Resolve conflicts if they cannot be resolved otherwise

Project Role: Group leader

- Responsible for intra-group communication (Meeting Management: Primary Facilitator)
 - Run the weekly project meeting
 - Post agenda before meeting
 - Define and keep track of action items (who, what, when)
 - Measure progress (Enforce milestones)
 - Deliver work packages for the tasks to the project management
 - Present problems and status of team to project manager
- The group leader has to be rotated among members of the team.

Group Leader: Create an Agenda

- Purpose of Meeting
- Desired Outcome
- Information Sharing
- Information Processing
- Meeting Critique

Action Items
(Check Previous
Meeting)

Issues
(Check Previous
Meeting & BBoards)

The screenshot shows a window titled "New Agenda" for the "Agenda for Database Group". It includes fields for "Date" (06/19/96), "Location", "Start Time" (12:00 PM), "End Time" (12:00 PM), "Primary Facilitator" (Bernd Bruegge), "Minute Taker", and "Time Keeper". Below these are sections for "Purpose of the Meeting", "2. Desired Outcome", "3. Information Sharing (15 Minutes)", and "4. Information Processing (40 Minutes)". Each of the last three sections has a toggle switch for "Include [Action Item/Issue] Text" (set to "Yes") and an "Update" button. Explanatory text below each toggle describes how to use the switch and the update button.

Agenda for Database Group

Date: 06/19/96

Location: Primary Facilitator: Bernd Bruegge

Start Time: 12:00 PM Minute Taker:

End Time: 12:00 PM Time Keeper:

Purpose of the Meeting

2. Desired Outcome

3. Information Sharing (15 Minutes)

Include Action Item Text: Yes Update Action Item Text

To exclude Action Items, choose 'No' and press 'Update Action Item Text'.
To include Action Items, choose 'Yes' and press 'Update Action Item Text'.
These two fields form a toggle switch for including Action Items in this Agenda.
The 'Update Action Item Text' button can also be used to refresh the Action Items in an Agenda.

4. Information Processing (40 Minutes)

Include Issue Text: Yes Update Issue Text

To exclude Issues, choose 'No' and press 'Update Issue Text'.
To include Issues, choose 'Yes' and press 'Update Issue Text'.
These two fields form a toggle switch for including Issues in this Agenda.
The 'Update Issue Text' button can also be used to refresh the set of Issues in an Agenda.

Project Role: Liaison

- Responsible for inter-group communication
 - Make available public definitions of subsystem developed by the team to the architecture teams (ensure consistency, etc)
 - Coordinate tasks spanning more than one group with other teams
- Responsible for team negotiations
- Examples: API Engineer, Configuration manager

Project Role: Planner

- Plans and tracks the activities of an individual team and has the following responsibilities.
- Define project plan for team:
 - PERT chart, resource table and GANTT chart showing work packages
 - Enter project plan into project management tool
- Make project plan available to management
- Report team status to project manager

No explicit planner in PAID. Responsibilities assumed by coaches

Project Role: Document Editor

- Collect, proofread and distribute team documentation
- Submit team documentation to architecture team
- Collect agendas
- Take minutes at meetings

Web Master

- Maintain team home page
- Keep track of meeting history
- Keep track of design rationale

Web Master

- Publish Meeting Information on Team Homepage
 - Must contain Agenda, minutes, action items and issues
 - Possibilities:
 - One HTML document per meeting, with anchors (maintained by one role)
 - Separate HTML documents for Agenda, Minutes, etc (maintained by several roles)

Date	Agenda	Minutes	Action Items	Issues
9/9/96	<u>Agenda</u>	<u>Minutes</u>	<u>Action Items</u>	<u>Issues</u>
9/16/96	<u>Agenda</u>	<u>Minutes</u>	<u>Action Items</u>	<u>Issues</u>

<http://cascade1.se.cs.cmu.edu/15-413/homePagesTeams/UserInterface/www/index.htm>

- Reference:
 - Bruegge&Dutoit, Chapter 12
 - <http://notesbruegge.in.tum.de/PAID2/schedule/ProjectManagement011599.pdf>
 - Cost estimation
 - Reference: Software engineering economics, Barry Boehm, Prentice Hall 1981